

TITLE PAGE

Problem Statement ID – SOAIDEATHON-H23

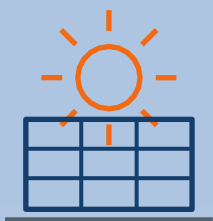
**Problem Statement Title – AI-Orchestrated
Campus Microgrid with Solar, Storage, EV and
Critical-Load Resilience**

Theme – Renewable / Sustainable Energy PS

Category – Hardware

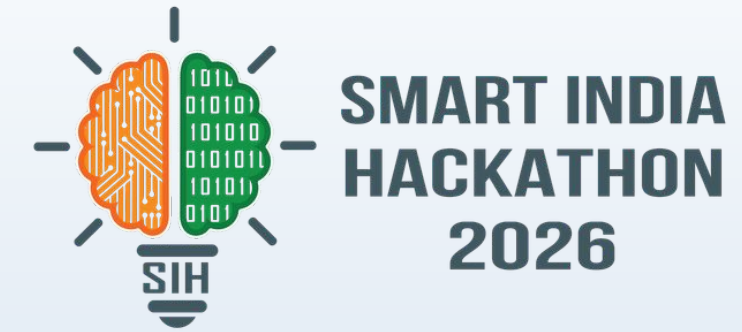
Team ID –

Team Name – Compile and Pray





IDEA TITLE



AI-Orchestrated Campus Microgrid with Solar, Storage, EV and Critical-Load Resilience

PROBLEM

- CAMPUS DEMAND AND GENERATION CHANGE THROUGHOUT THE DAY
- CONVENTIONAL MICROGRIDS SUFFER FROM UNCOORDINATED LOAD DISTRIBUTION
- WEATHER CONDITIONS CAN CHANGE ANY TIME LEADS TO CRITICAL POWER OUTAGES

PROPOSED SOLUTION

- EDGE CONNECTED ESP-32 TELEMETRY RELAYS REALTIME VOLTAGE/CURRENT DATA TO BACKEND SERVER
- FLASK ENGINE ENGINE CONTINUOUSLY BALANCES LOADS

UNIQUE INNOVATION

- INTEGRATES OPEN METEO LIVE API FORECAST
- ONSITE RAIN AND TEMPERATURE SENSORS
- PREDICTIVE LOAD SHEDDING

TECHNICAL APPROACH

HARDWARE LAYER

- **ESP32 MICROCONTROLLER:**
 - . WIFI HTTP TELEMETRY AND RELAY DRIVER
- **INA3221 SENSORS**
 - . MULTICHANNEL V/I POWER PROFILING
- **OTHER SENSORS**
 - . LM35 TEMP SENSOR
 - . RAIN SENSOR
- **DISPLAY**
 - . SSD1306 OLED LOCAL DASHBOARD

HARDWARE LAYER

- **FLASK MICRO-FRAMEWORK**
 - . REST API ENDPOINTS
- **SQLite DATABASE**
 - . TELEMETRY HISTORY LOGGING
- **OPEN METEO API**
 - . UV-INDEX
 - . RAIN/CLOUD FORECAST
- **CORS HANDLING**
 - . REALTIME DASHBOARD STREAM

PROCESS METHODOLOGY

- **DATA INGESTION**
 - . ESP32 POSTS JSON VIA SENSOR
- **WEATHER SYNC**
 - . 10 SECOND CACHE AUTO REFRESH
- **AI EVALUATION**
 - . RULE BASED HEURISTIC LOAD PRIORITIZATION
 - . RAIN/CLOUD FORECAST
- **RELAY ACTION**
 - . DYNAMIC L1-L4 LINE SHED/RESTORE

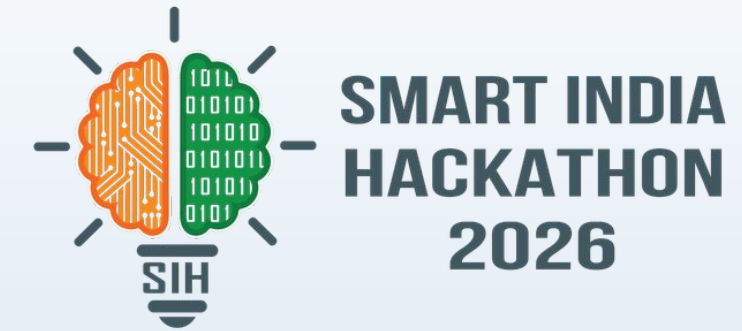
IMPLEMENTATION FLOW



Prototype output: live energy dashboard + recommended dispatch schedule + outage mode + explanation panel + manual override.



FEASIBILITY AND VIABILITY



FEASIBILITY

TESTED WORKING PROTOTYPE VALIDATES OPERATIONAL VIABILITY:

- **LOW HARDWARE COST:**
 - BUILT USING MODULAR, OFF-THE-SHELF ESP32 AND INA3221 COMPONENTS
- **LIGHTWEIGHT STACK:**
 - SQLITE + FLASK RUNS EFFICIENTLY ON EDGE GATEWAYS (E.G. RASPBERRY PI)
- **PROVEN INTEROPERABILITY:**
 - SEAMLESS HTTP JSON PROTOCOL BRIDGES MICROCONTROLLERS TO WEB BACKEND

CHALLENGES / RISKS

IDENTIFIED OPERATIONAL AND ENVIRONMENTAL VULNERABILITIES:

- **NETWORK DROPOUT:**
 - DISRUPTION IN WI-FI CAUSING TELEMETRY UPLOAD DELAYS
- **WEATHER API LATENCY:**
 - STALE WEATHER FORECASTS DURING SUDDEN MICRO-CLIMATES
- **OVERCURRENT RISK:**
 - RAPID LOAD SPIKES OVERWHELMING BATTERY STORAGE

MITIGATION

ENGINEERING SAFEGUARDS EMBEDDED INTO SOFTWARE & HARDWARE:

- **LOCAL EDGE FAILOVER:**
 - ESP32 OPERATES STANDALONE RULES WHEN SERVER CONNECTION DROPS
- **HYBRID SENSING:**
 - ON-BOARD RAIN SENSOR OVERRIDES API IF WEATHER SUDDENLY TURNS
- **HARDWARE FAILSAFE:**
 - DEDICATED ANALOG CUTOFF LIMITS ON MOTOR & BACKUP CHANNELS

TARGET AUDIENCE

- UNIVERSITY / CAMPUS FACILITY MANAGERS
- ENERGY MANAGERS AND MICROGRID OPERATORS
- CRITICAL CAMPUS FACILITIES: LABS, DATA ROOMS, HEALTH/SAFETY SYSTEMS
- EV CHARGING INFRASTRUCTURE OPERATORS
- STUDENTS AND STAFF WHO DEPEND ON RELIABLE CAMPUS SERVICES

EXPECTED BENEFITS

- ENVIRONMENTAL IMPACT
MAXIMIZES LOCALIZED SOLAR UTILIZATION;
REDUCES RELIANCE ON DIESEL GENERATORS AND
FOSSIL FUEL GRID DRAWDOWN BY UP TO 35%.
- ECONOMIC BENEFITS
PREVENTS COSTLY BATTERY DEEP-DISCHARGES
AND LOWERS OPERATIONAL ELECTRICITY BILLS
THROUGH DYNAMIC PEAK-SHAVING.
- SOCIAL VALUE
ENSURES UNINTERRUPTIBLE POWER FOR ESSENTIAL
LOADS (LIGHTING, MEDICAL/USB DEVICES) IN RURAL
AND REMOTE MICROGRIDS.

**Success metrics: forecast error • renewable utilization • peak-demand reduction • critical-load survival time
• battery SoC compliance • EV charging completion • operator override rate**

RESEARCH AND REFERENCES

RESEARCH BASIS

- DOE describes microgrids as controllable local systems that can operate grid-connected or independently during outages.
- DOE/NREL research highlights forecasting, advanced control, storage and EV-grid integration as important parts of modern microgrid operation.
- Our prototype applies these concepts at campus scale with explainable scheduling and operator override.

REFERENCES / LINKS

1. U.S. Department of Energy - Microgrid Systems: <https://www.energy.gov/oe/microgrid-systems>
2. U.S. Department of Energy - Microgrids: <https://www.energy.gov/indianenergy/tribal-energy-guide/microgrids>
3. NREL - Solar and Wind Forecasting: <https://www.nrel.gov/grid/solar-wind-forecasting>
4. NREL - EV-Grid Integration Control & System Implementation: <https://research-hub.nrel.gov/en/publications/ev-grid-integration-evgi-control-and-system-implementation-resear/>
5. DOE - Solar Integration: DER and Microgrids Basics: <https://www.energy.gov/cmei/systems/solar-integration-distributed-energy-resources-and-microgrids-basics>